

## TALARA, Peru

WMO: 843900

Lat: 4.577S

Lon: 81.254W

Elev: 86

StdP: 100.30

Time Zone: -5.00 (W05)

Period: 90-00

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 8             | 15.8       | 15.9 | 13.3                          | 9.6 | 16.6 | 13.8 | 9.9 | 17.0 | 18.6                  | 18.8 | 15.5 | 19.1 | 5.6                   | 170  | 0.485 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 7.5                    | 31.9            | 24.4 | 31.1 | 24.1 | 30.5 | 23.8 | 26.3                | 29.1 | 25.9 | 28.7 | 25.3 | 28.2 | 4.9                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.8                            | 21.3 | 28.3 | 25.1 | 20.4 | 27.6 | 24.2 | 19.4 | 27.1 | 82.9          | 29.3 | 80.5 | 28.9 | 77.7 | 27.6 | 28.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
| Min               | Max  |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 15.3              | 13.2 | 10.5 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |
|                   |      |      | 13.3                       | 26.5 | 2.1                | 1.4 | 11.8                                               | 27.5 | 10.5       | 28.3 | 9.4        | 29.1 | 7.9        | 30.1 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 23.7   | 25.7  | 27.0  | 27.0  | 26.0  | 24.9  | 22.8  | 21.5  | 20.9  | 21.1  | 21.4  | 22.4  | 24.1  |
|                                                                        | DBStd       | 2.72   | 1.36  | 1.05  | 1.13  | 1.32  | 1.64  | 1.81  | 1.88  | 1.70  | 1.73  | 1.54  | 1.75  | 1.81  |
|                                                                        | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD18.3     | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDD10.0     | 5002   | 487   | 474   | 527   | 478   | 461   | 385   | 356   | 337   | 333   | 354   | 373   | 436   |
|                                                                        | CDD18.3     | 1961   | 229   | 241   | 268   | 228   | 202   | 135   | 98    | 79    | 83    | 96    | 123   | 178   |
|                                                                        | CDH23.3     | 11036  | 1512  | 1933  | 2170  | 1535  | 1136  | 487   | 258   | 171   | 203   | 241   | 451   | 940   |
|                                                                        | CDH26.7     | 2887   | 437   | 607   | 676   | 452   | 272   | 67    | 31    | 18    | 29    | 26    | 69    | 203   |
| (14) Wind                                                              | WSAvg       | 5.4    | 5.3   | 4.4   | 4.0   | 5.1   | 5.7   | 6.1   | 6.1   | 5.9   | 5.5   | 5.9   | 5.7   | 5.5   |
| (15) Precipitation                                                     | PrecAvg     | 10     | 1     | 3     | 7     | 2     | 1     | 1     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | PrecMax     | 133    | 32    | 37    | 122   | 40    | 41    | 22    | 0     | 0     | 0     | 8     | 2     | 1     |
|                                                                        | PrecMin     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | PrecStd     | 29     | 6     | 7     | 21    | 7     | 7     | 4     | 0     | 0     | 0     | 1     | 0     | 0     |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 32.0  | 32.6  | 32.9  | 32.0  | 31.2  | 29.8  | 29.0  | 28.2  | 29.1  | 28.8  | 30.2  | 31.2  |
|                                                                        |             | MCWB   | 24.6  | 24.2  | 24.7  | 24.5  | 24.1  | 23.2  | 22.9  | 22.3  | 23.1  | 22.1  | 23.5  | 25.0  |
|                                                                        | 2%          | DB     | 30.9  | 31.8  | 31.9  | 31.0  | 30.1  | 28.2  | 27.4  | 27.0  | 27.1  | 27.2  | 28.9  | 30.1  |
|                                                                        |             | MCWB   | 23.8  | 24.1  | 24.3  | 24.1  | 23.4  | 22.1  | 21.7  | 21.6  | 22.0  | 21.2  | 22.8  | 24.2  |
|                                                                        | 5%          | DB     | 30.0  | 31.0  | 31.0  | 30.2  | 29.2  | 27.2  | 25.9  | 25.0  | 25.1  | 25.8  | 26.9  | 28.9  |
|                                                                        |             | MCWB   | 23.3  | 23.9  | 24.1  | 23.6  | 22.9  | 21.5  | 20.6  | 20.2  | 20.2  | 20.3  | 21.1  | 22.7  |
|                                                                        | 10%         | DB     | 29.1  | 30.1  | 30.1  | 29.5  | 28.1  | 26.1  | 24.8  | 23.9  | 24.1  | 24.7  | 25.8  | 27.8  |
|                                                                        |             | MCWB   | 22.9  | 23.6  | 24.0  | 23.3  | 22.2  | 21.0  | 20.0  | 19.4  | 19.4  | 19.7  | 20.4  | 22.0  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 26.6  | 27.1  | 27.0  | 26.2  | 25.0  | 23.6  | 23.0  | 22.7  | 23.4  | 22.5  | 24.0  | 26.1  |
|                                                                        |             | MCDB   | 29.7  | 29.7  | 29.9  | 29.4  | 30.0  | 28.7  | 28.5  | 27.6  | 28.7  | 27.6  | 29.4  | 30.0  |
|                                                                        | 2%          | WB     | 25.9  | 26.4  | 26.2  | 25.4  | 24.1  | 22.7  | 22.2  | 21.9  | 22.4  | 21.5  | 23.0  | 25.1  |
|                                                                        |             | MCDB   | 28.6  | 28.7  | 29.0  | 28.8  | 29.1  | 27.0  | 26.4  | 26.4  | 26.6  | 26.1  | 28.0  | 28.7  |
|                                                                        | 5%          | WB     | 25.1  | 25.8  | 25.7  | 24.9  | 23.4  | 22.2  | 21.3  | 20.9  | 21.1  | 20.8  | 22.2  | 24.2  |
|                                                                        |             | MCDB   | 27.4  | 28.1  | 28.6  | 28.5  | 27.9  | 26.0  | 24.7  | 24.1  | 23.0  | 25.0  | 25.7  | 27.2  |
|                                                                        | 10%         | WB     | 24.1  | 24.8  | 25.0  | 24.1  | 22.9  | 21.5  | 20.5  | 20.0  | 20.2  | 20.0  | 21.1  | 23.0  |
|                                                                        |             | MCDB   | 27.5  | 27.8  | 28.1  | 27.8  | 26.9  | 25.4  | 24.1  | 23.3  | 23.9  | 23.9  | 25.1  | 26.3  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 8.1   | 7.5   | 7.4   | 8.0   | 7.9   | 7.1   | 7.1   | 7.1   | 7.5   | 7.6   | 7.7   | 8.0   |
|                                                                        |             | MCDBR  | 8.3   | 8.1   | 8.3   | 8.3   | 8.3   | 7.7   | 7.7   | 7.7   | 8.1   | 8.1   | 8.2   | 8.4   |
|                                                                        |             | MCWBR  | 3.4   | 3.3   | 3.5   | 3.6   | 3.7   | 3.5   | 3.4   | 3.8   | 3.8   | 3.7   | 3.6   | 3.7   |
|                                                                        | 5% WB       | MCDBR  | 6.6   | 5.8   | 6.4   | 7.0   | 7.8   | 7.3   | 7.5   | 7.5   | 7.8   | 7.9   | 7.8   | 7.1   |
|                                                                        |             | MCWBR  | 3.2   | 3.0   | 3.1   | 3.2   | 3.7   | 3.5   | 3.5   | 3.7   | 3.6   | 3.7   | 3.4   | 3.5   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.430 | 0.423 | 0.422 | 0.448 | 0.461 | 0.457 | 0.459 | 0.488 | 0.514 | 0.459 | 0.458 | 0.454 |
|                                                                        | taud        |        | 2.396 | 2.427 | 2.433 | 2.355 | 2.305 | 2.314 | 2.311 | 2.199 | 2.114 | 2.293 | 2.280 | 2.308 |
|                                                                        | Ebn at Noon |        | 907   | 913   | 903   | 855   | 817   | 805   | 811   | 809   | 811   | 870   | 875   | 881   |
|                                                                        | Edh at Noon |        | 126   | 123   | 121   | 126   | 127   | 123   | 125   | 145   | 164   | 139   | 141   | 136   |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 6.05  | 5.83  | 6.02  | 6.08  | 5.51  | 5.03  | 5.02  | 5.41  | 5.98  | 6.25  | 6.43  | 6.32  |
|                                                                        | RadStd      |        | 0.40  | 0.24  | 0.36  | 0.20  | 0.29  | 0.33  | 0.31  | 0.35  | 0.30  | 0.24  | 0.28  | 0.25  |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (2 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page